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EBU5608 Product Development and Management

Topic 14 – Introduction to Management

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Agenda

- **Management**
  - Definition
  - The 3Es – Efficiency, Effectiveness and Equity
  - The roles of a manager

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The Big Question

- Why do you think engineers should study management?

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The Big Question

You need to study this to

- Understand the basic **concepts** and principles of management
- Build your business **vocabulary**
- Develop your workplace **skills**
- Learn about various forms and functions of technology **organisations**
- Appreciate today's business **careers**

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Management and managers

- What **is** management?
- What do managers **do**?
- What **skills**, **styles** and **capabilities** do they need?
- How do managers **learn** and develop?
- Why **study** management as a discipline?
- Can management **theories** enhance learning and enlighten **practice**?

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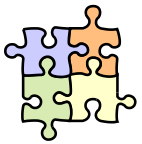
Management: a process

Management is a process of achieving organisational **objectives** within a changing environment, by **balancing** efficiency, effectiveness and equity, obtaining the **most** from limited resources, and working with and through **people.**

Let's look at those a little more…



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Management: a process - objectives and balance

1. Achieving organisational **objectives** within a changing **environment**
   - An objective is a **target** or aim to be striven for
   - In this course, you'll find out that an organisation, or an individual, is more successful if they seek outcomes that are both **challenging** and **achievable**
   - Managers, especially senior managers, spend a great deal of their time, learning about the changing **environment**
2. **Balancing** efficiency, effectiveness and equity
   - we'll look at these later

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Management: a process - resources & people

3. Obtaining the most from limited **resources**
   - Recognizing that resources are limited is more than a question of using them most **efficiently**
   - Where materials are scarce, **innovation** may be needed
   - This means **adapting** products or processes to use fewer or alternative resources
4. Working with and through **people**
   - Management is primarily a **social** process, often defined as 'getting things done through people'
   - There's a danger of presenting management as a set of **techniques**, such as for optimising plans, but eventually, **people** put all plans into effect

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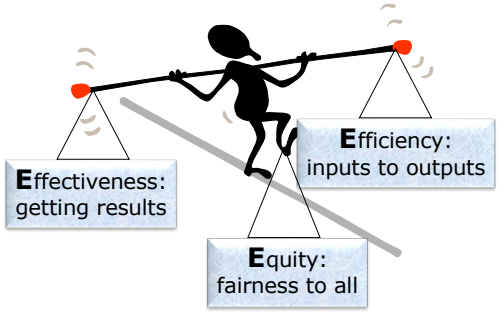
Balancing the 3 Es



Trying to achieve 3 Es of the management balance is a dilemma. Success against one criterion is often at the expense of another

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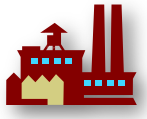
The 3 Es - Efficiency

- **Efficiency** is a measure of how well input resources are transformed into outputs
- For example:
  - Washing machine suppliers state that their washing machine can save energy
  - By using less water/electricity, their washing machine can wash more loads than other brands for the same cost
- Similarly, a more efficient **factory** can produce more goods from less labour and/or raw material



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The 3 Es - Effectiveness

- **Effectiveness** is an assessment of how far a stated **objective** is achieved
- Often, a focus on 'getting the job done' is important, e.g. when managing a **crisis**
- War, the ultimate crisis, requires leaders who believe in 'winning at all costs'
- But even in such circumstances, good generals consider how to **conserve** and **deploy** their resources

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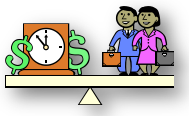
The 3 Es – Effectiveness or Efficiency?

- A leader's **overemphasis** on effectiveness leads to a **loss** of efficiency
  - Resources are **wasted**
- On the other hand, **too much** stress on efficiency may mean that the task may **not** get done at all
- The right **balance** is a management decision



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The 3 Es - Equity

- **Equity** is the third ingredient of the mixture of Es
- This concerns the **fair distribution** of outputs among recipients
- Many **public** and **not-for-profit** organisations place equity at the core of their objectives
- Can you think of examples?

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Management, Technology and Innovation

- Management plays a critical role in the development and commercialization of new technologies
- Management professionals work to identify market opportunities, secure funding, and develop business plans that enable new technologies to be brought to market
- Technology management is a field of study and practice that combines management principles with technology-related knowledge and expertise
- It focuses on planning, developing, implementing, and using technology to improve business operations and achieve strategic objectives

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Management, Technology and Innovation

- The use of technology has allowed businesses to automate repetitive tasks, manage data more efficiently, communicate with their customers and employees more effectively, and innovate to meet the changing needs of customers

Examples

- Collaboration tools such as **Slack**, **Microsoft Teams**, and **Asana** facilitate real-time communication, file sharing, and task tracking
- Project management tools like **Trello**, **Jira**, and **Monday.com** offer advanced features for project planning, task assignment, and progress tracking
- Cloud-based tools such as **Microsoft Project Online**, **Smartsheet**, and **Wrike** allow teams to store, share, and collaborate on project documents in real-time
- The integration of artificial intelligence and automation in project management has significantly improved efficiency – AI algorithms can analyze historical project data to provide insights into potential risks and opportunities

Business vs. non-profit organisations

Business

- An individual or organization that tries to earn a profit by providing products that satisfy people's needs
- Examples are IBM and Coca-Cola

Non-profit organisation

- Provides products, especially services, for some purpose other than profits
- Examples are
  - Religious/social – such as The Salvation Army
  - Colleges & universities – such as QMUL

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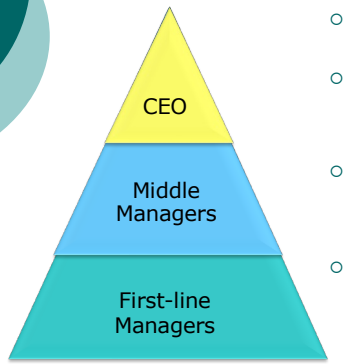
Looking at managers - management levels



- There are 3 basic management levels in most organisations
- Top-level management
  - president, executive vice president
- Middle managers
  - chief engineer, division head etc
- First-line managers
  - foreman, supervisor, section chief

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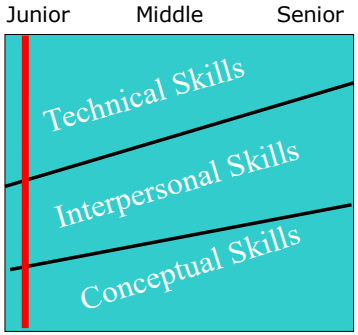
Looking at managers - skills



Management Skills Composition

- The higher you move up in management
  - less technical skills are required
  - but more interpersonal and conceptual skills are needed

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Managerial Roles (What Managers Do)

- The roles of managers fall into 3 groups
  - Interpersonal
  - Informational
  - Decisional

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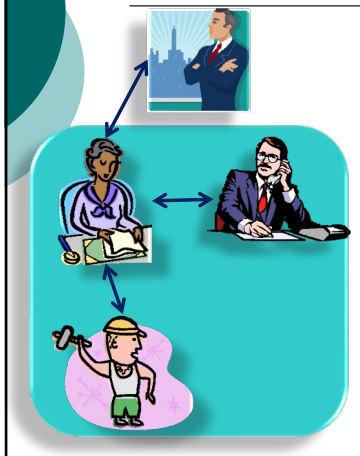
Managerial roles - interpersonal role



- **Figurehead** role
  - Handles outward relationships
  - Embodies legal authority; leads formal duties, e.g. signing documents, accepting visitors
- **Leader** role
  - Downward relationships
  - Motivates others to do their jobs
- **Liaison** role
  - Horizontal relationships in company

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Managerial roles - informational role

- **Monitor** Role
  - Collects information about internal operations and external events

- **Disseminator** Role
  - Transforms information internally to everybody in organization (like a telephone switchboard)

- **Spokesman** Role
  - Public relations

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Managerial roles - decisional role

- **Entrepreneurial** Role
  - Initiates changes, assumes risks, transforms ideas into useful products
- **Disturbance Handler** Role
  - Deals with unforeseen problems and crises
- **Resource Allocator** Role
  - Distributing resources
- **Negotiator** Role
  - Bargains with suppliers, customers etc in favour of their enterprise

Launching a New 5G Smartphone at Huawei

- Huawei is preparing to launch a new **5G-enabled smartphone** that aims to outperform its competitors by offering **superior network connectivity**, **advanced camera technology**, and a **longer battery life**. The product is targeted to both **international and domestic markets**, with a focus on **innovation**, **cost efficiency**, and **meeting stringent quality standards**.
- As part of the **product development team**, you are an **engineering manager** overseeing the **hardware design**, **software integration**, and **production processes**. The goal is to launch the product in 6 months, but the project is facing several challenges, including **supply chain delays**, **increasing production costs**, and **intense competition** from other smartphone makers, particularly in the 5G space.

Key Managerial Skills and Their Importance

- Discuss these key managerial skills, how they apply to the scenario above and why they are important
- *Project Management & Time Management*
- *Cross-Functional Team Coordination & Communication*
- *Problem-Solving & Risk Management*
- *Leadership and People Management*
- *Budgeting and Financial Management*
- *Decision-Making and Adaptability*

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Summary

- 'Management is a process of achieving organisational **objectives** within a changing **environment**, by balancing **efficiency**, **effectiveness** and **equity**, obtaining the most from **limited resources**, and working with and through **people**.
- There are 3 basic management levels in most organisations
  - Top-level management
  - Middle managers
  - First-line managers
- The roles of managers fall into 3 groups
  - Interpersonal
  - Informational
  - Decisional

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In-class Exercise

- Work in groups of 3-4 students.
- You are tasked with managing a new AI startup focused on developing a product that uses AI for automated engineering design (e.g., AI software that creates product designs based on input parameters). The startup aims to bring innovation to the engineering industry.
- Consider how management principles – especially the balance between efficiency, effectiveness, and equity – apply to a tech startup.

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In-class Exercise

- **Efficiency**: How will the startup use resources (time, money, people) to maximize productivity? **Effectiveness**: How will the startup ensure they meet their business goals, such as launching the product on time or gaining market traction?
- **Equity**: How will the startup ensure fairness in hiring practices, resource distribution, and offering equal opportunities to employees and customers?

- Discuss in your group:
  - What strategies will the group use to balance these 3 Es in their startup?
  - How could AI and technology help the startup achieve these goals?

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Recommended Reading

1. *Management*, S. P. Robbins and M.A. Coulter, 14th Edition, 2018, Pearson, Chapter 1

References

1. Adapted from *Management*, S. P. Robbins and M.A. Coulter, 14th Edition, 2018, Pearson, Chapter 1
2. Adapted from *Management*, S. P. Robbins and M.A. Coulter, 14th Edition, 2018, Pearson, Chapter 1
3. Diagram taken from Palmer, pg.4
4. http://miranda.sourceoecd.org/vl=312866/cl=35/nw=1/rpsv/factbook/10-03-02.htm
5. Adapted from e-Business Technologies, Craig Van Slyke, France Belanger,2003, Wiley, pg. 25
6. .. Pg. 24
7. Mintzberg, H (1989), *Mintzberg on Management*, Free Press, New York, NY.